Math 110 — Week 5

Solutions

HW5 P1.

Idea. The null space is given by two zeros, so factor. Surjectivity: find polynomials sent to e_1 and e_2. Those same polynomials give the basis of the quotient.

1. (a) $p \in \operatorname{null} T$ means $p(0) = p(1) = 0$, so $x(x-1)$ divides p (W7 of the handout, applied at 0 and then at 1). With $\deg p \le 3$,
$$p = x(x-1)(ax+b) = a\,x^2(x-1) + b\,x(x-1).$$
So $\operatorname{null} T \subseteq \operatorname{span}\big(x(x-1),\ x^2(x-1)\big)$; conversely both polynomials vanish at 0 and 1, so the span is in $\operatorname{null} T$. They have different degrees, so they are independent: $\dim \operatorname{null} T = 2$.
2. $T(1-x) = (1,0)$ and $T(x) = (0,1)$. These span $\mathbb{R}^2$, so T is surjective. (Check: $4 = 2+2$.)
3. (b) $\widetilde{T}\colon \mathcal{P}_3(\mathbb{R})/\operatorname{null} T \to \mathbb{R}^2$, $\widetilde{T}(p + \operatorname{null} T) = (p(0), p(1))$. It is well-defined because two names of the same set differ by a polynomial vanishing at 0 and 1.
4. (c) $\widetilde{T}$ is an isomorphism, and $\widetilde{T}\big((1-x) + \operatorname{null} T\big) = e_1$, $\widetilde{T}\big(x + \operatorname{null} T\big) = e_2$. The inverse of an isomorphism sends a basis to a basis, so $(1-x) + \operatorname{null} T,\ x + \operatorname{null} T$ is a basis of the quotient.

HW5 P2.

Idea. The formula is forced, so uniqueness is immediate. Existence is the well-definedness check, and it uses exactly $U \subseteq \operatorname{null} T$.

1. *Uniqueness.* Every element of V/U is $v + U$ for some v, and the condition prescribes the value Tv there. So at most one such map exists.
2. *Well-defined.* If $v + U = w + U$, then $v - w \in U \subseteq \operatorname{null} T$, so $Tv - Tw = T(v-w) = 0$.
3. *Linear.*
$$\widetilde{T}(\lambda(v+U) + (w+U)) = \widetilde{T}((\lambda v + w) + U) = T(\lambda v + w) = \lambda\widetilde{T}(v+U) + \widetilde{T}(w+U).$$
4. *Null space.* $\widetilde{T}(v+U) = 0 \iff Tv = 0 \iff v \in \operatorname{null} T$. So $\operatorname{null}\widetilde{T} = \{v + U : v \in \operatorname{null} T\} = (\operatorname{null} T)/U$.

HW5 P3.

Idea. If φ takes one nonzero value, scaling the input produces every value.

1. Suppose $\varphi \neq 0$. Choose v with $\varphi(v) = c \neq 0$.
2. For any $a \in \mathbb{F}$:
$$\varphi\big(\frac{a}{c}v\big) = \frac{a}{c}\varphi(v) = a.$$
So φ is surjective.

HW5 P4.

Idea. φ_j reads off the j-th coordinate. Functionals are compared by evaluating on the basis.

1. (a) Write $v = \sum_k a_k v_k$. Then $\varphi_j(v) = \sum_k a_k \varphi_j(v_k) = a_j$. Substituting $a_j = \varphi_j(v)$ gives the formula.
2. (b) Evaluate the right side at v_k: $\sum_j \psi(v_j)\varphi_j(v_k) = \psi(v_k)$. So the two functionals ψ and $\sum_j \psi(v_j)\varphi_j$ agree on a basis of V, hence are equal.

HW5 P5.

Idea. Column j of $\mathcal{M}(T')$ is the coordinate vector of $T'(w_j')$, and coordinates of a functional are its values on $v_1, \dots, v_n$.

1. Let $B = (v_1, \dots, v_n)$, $C = (w_1, \dots, w_m)$, $A = (a_{lk})$, so $Tv_k = \sum_l a_{lk} w_l$.
2. Evaluate $T'(w_j')$ on the basis vector v_k, using column k of $\mathcal{M}(T)$:
$$T'(w_j')(v_k) = w_j'(Tv_k) = w_j'\Big(\sum_l a_{lk} w_l\Big) = a_{jk}.$$
3. By P4(b),
$$T'(w_j') = \sum_k T'(w_j')(v_k)\, v_k' = \sum_k a_{jk} v_k'.$$
4. So the (k, j) entry of $\mathcal{M}(T')$ is a_{jk}: $\mathcal{M}(T') = A^{\mathsf{T}}$.

HW5 P6.

Idea. A row is in U^0 when it gives 0 on each of the two spanning vectors.

1. (a) $(a\ b\ c)$ applied to $(1, 1, 0)$ is $a + b$; applied to $(0, 1, 1)$ is $b + c$. Both are zero exactly when $a = -b$ and $c = -b$.
2. $U^0 = \operatorname{span}\big((1\ \ -1\ \ 1)\big)$, the functional $(x, y, z) \mapsto x - y + z$.
3. (b) $\dim U = 2$ (the two vectors are not multiples of each other), $\dim U^0 = 1$, and $2 + 1 = 3$.

HW5 P7.

Idea. Injective: for $v \neq 0$ build a functional with $\varphi(v) = 1$ by making v a basis vector.

1. j_V is linear:
$$j_V(\lambda v + w)(\varphi) = \varphi(\lambda v + w) = \lambda\varphi(v) + \varphi(w).$$
2. Let $v \neq 0$. Extend v to a basis of V and let φ be 1 on v and 0 on the other basis vectors. Then $j_V(v)(\varphi) = 1$, so $j_V(v) \neq 0$. Hence $\operatorname{null} j_V = \{0\}$.
3. If $\dim V = n < \infty$: $\dim V'' = \dim V' = n$. An injective map between spaces of the same finite dimension is an isomorphism.

R1.

Idea. E is the map of HW5 P1, so the null space comes from factoring out $x(x-1)$, and the quotient basis from two polynomials sent to e_1, e_2; the isomorphism is the first isomorphism theorem checked by hand.

1. (a) E is linear:
$$E(\lambda p + q) = (\lambda p(0) + q(0), \lambda p(1) + q(1)) = \lambda E(p) + E(q).$$
$E(1 - x) = (1, 0)$ and $E(x) = (0, 1)$, so $\operatorname{range} E$ contains a basis of $\mathbb{R}^2$: E is surjective.
2. (b) $p \in U$ means $p(0) = p(1) = 0$, so $p = x(x-1)(ax + b)$ (W7, twice). So $U = \operatorname{span}(x(x-1),\ x^2(x-1))$; the two have different degrees, so they are independent, and they are a basis of U ($\dim U = 4 - 2 = 2$ by rank–nullity).
3. (c) Well defined: if $p + U = q + U$, then $p - q \in U = \operatorname{null} E$, so $E(p) = E(q)$. Linear because E is. Injective: if $\widetilde{E}(p + U) = 0$, then $p \in U$, so $p + U = 0 + U$. Surjective because E is.
4. (d) $(1 - x) + U \mapsto (1, 0)$ and $x + U \mapsto (0, 1)$. These two elements form a basis of V/U, because their images form a basis and $\widetilde{E}$ is an isomorphism (W3).

R2.

Idea. Everything follows from $(T^*\varphi)(v) = \varphi(Tv)$.

1. (a) $T^*(\varphi) = \varphi \circ T$ for $\varphi \in W^*$. It is a functional on V, being a composition of linear maps. T^* is linear:
$$T^*(\lambda\varphi + \psi) = (\lambda\varphi + \psi) \circ T = \lambda(\varphi \circ T) + \psi \circ T = \lambda T^*\varphi + T^*\psi.$$
2. (b) For $\varphi \in Z^*$:
$$(S \circ T)^*\varphi = \varphi \circ S \circ T = (S^*\varphi) \circ T = T^*(S^*\varphi).$$
3. (c) Suppose $T^*\varphi = 0$, that is, $\varphi(Tv) = 0$ for all v. Since T is surjective, every $w \in W$ is some Tv, so $\varphi(w) = 0$ for all w. So $\varphi = 0$, and T^* is injective.
4. (d) $(T^*\ell)(x, y) = \ell(2x - y) = 6x - 3y$.

R3.

Idea. Membership in U^0 is two linear equations. A dual basis of a basis that starts with a basis of U has its last element in U^0 automatically.

1. (a) $U^0 = \{\varphi \in V' : \varphi(u) = 0 \text{ for all } u \in U\}$, the functionals on $\mathbb{R}^3$ that vanish on U.
2. (b) A row $(a\ b\ c)$ is in U^0 exactly when $a + b = 0$ and $b + c = 0$ (it is enough to check the spanning vectors). So $(a\ b\ c) = b(-1\ \ 1\ \ -1)$, and $(1\ -1\ 1)$ is a basis of U^0.
3. (c) Add $(1, 0, 1)$: the matrix with columns $(1, 1, 0), (0, 1, 1), (1, 0, 1)$ has determinant $2 \neq 0$, so B is a basis. The dual basis is the rows of the inverse:
$$M = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \qquad M^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}.$$
So the dual basis is $\frac{1}{2}(1\ 1\ -1)$, $\frac{1}{2}(-1\ 1\ 1)$, $\frac{1}{2}(1\ -1\ 1)$.
4. (d) The third dual vector φ_3 has $\varphi_3(b_1) = \varphi_3(b_2) = 0$, and b_1, b_2 span U, so $\varphi_3 \in U^0$. It is nonzero, and $\dim U^0 = 3 - 2 = 1$ (LADR 3.125), so φ_3 spans U^0.

R4.

Idea. A functional on V is determined by its values on the basis $1, \sin x, \sin 2x$. Put those values in a matrix; the dual basis comes from its inverse.

1. The values $\delta_i(b_j)$, with rows $\delta_0, \delta_{\pi/2}, \delta_0'$ and columns $1, \sin x, \sin 2x$ (note $\sin\pi = 0$ and $(\sin 2x)' = 2\cos 2x$):
$$N = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}, \qquad N^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1/2 & -1/2 & 1/2 \end{bmatrix}.$$
2. (a) Row i of N is the coordinate vector of δ_i in the dual basis $\varphi_1, \varphi_2, \varphi_3$ (HW5 P4(b)). N is invertible, so the three rows are independent, and three independent vectors in the 3-dimensional V' are a basis (W1).
3. (b) If $\varphi_i = \sum_k c_{ik}\delta_k$, then $\varphi_i(b_j) = \sum_k c_{ik}N_{kj}$, and this must be δ_{ij}: $CN = I$, so $C = N^{-1}$. Reading the rows:
$$\varphi_1 = \delta_0, \qquad \varphi_2 = -\delta_0 + \delta_{\pi/2}, \qquad \varphi_3 = \tfrac{1}{2}\delta_0 - \tfrac{1}{2}\delta_{\pi/2} + \tfrac{1}{2}\delta_0'.$$
Check φ_3: on 1 it gives $\frac{1}{2} - \frac{1}{2} = 0$; on $\sin x$, $0 - \frac{1}{2} + \frac{1}{2} = 0$; on $\sin 2x$, $0 - 0 + \frac{1}{2} \cdot 2 = 1$.
4. (c) The values of φ on the basis are
$$\varphi(1) = 1, \qquad \varphi(\sin x) = \frac{1}{\pi}\int_0^\pi \sin x\, dx = \frac{2}{\pi}, \qquad \varphi(\sin 2x) = \frac{1}{\pi}\int_0^\pi \sin 2x\, dx = 0.$$
By HW5 P4(b), $\varphi = \varphi_1 + \frac{2}{\pi}\varphi_2$, and by (b)
$$\varphi = \delta_0 + \tfrac{2}{\pi}(-\delta_0 + \delta_{\pi/2}) = \big(1 - \tfrac{2}{\pi}\big)\delta_0 + \tfrac{2}{\pi}\delta_{\pi/2}.$$

R5.

Idea. Prop. 50 computes the null space and range of any dual map; Prop. 51 gives injectivity and surjectivity. Apply both to i and to π.

1. (a) $i'\varphi = \varphi \circ i$ sends $u \in U$ to $\varphi(u)$, so $i'\varphi = \varphi|_U$. i is injective, so i' is surjective (Prop. 51). $\operatorname{null} i' = (\operatorname{range} i)^0 = U^0$ (Prop. 50(i)).
2. (b) π is surjective, so π' is injective (Prop. 51). $\operatorname{range} \pi' = (\operatorname{null} \pi)^0 = U^0$ (Prop. 50(ii)).
3. (c) By the first isomorphism theorem for i', $V'/\operatorname{null} i' \cong \operatorname{range} i'$, that is, $V'/U^0 \cong U'$. By (b), π' is an isomorphism $(V/U)' \to U^0$.

R6.

Idea. Prop. 51 says which property of T matches which property of T': injective with surjective, surjective with injective.

1. (a) False. $T\colon \mathbb{R} \to \mathbb{R}^2$, $T(t) = (t, 0)$, is injective. For the row $\varphi = (0\ 1)$, $(T'\varphi)(t) = \varphi(t, 0) = 0$, so $T'\varphi = 0$ with $\varphi \neq 0$: T' is not injective.
2. (b) True: R2(c).
3. (c) True: $\dim U^0 = \dim V - \dim U$ (LADR 3.125) and $\dim(V/U) = \dim V - \dim U$ (Prop. 40).

E1.

Idea. Rows of the inverse of the matrix whose columns are the basis.

1. The matrix M with columns v_1, v_2 has $\det M = 1$, so
$$M = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad M^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$
2. The rows of M^{-1} are the dual basis: $M^{-1}M = I$ says that row i of M^{-1} applied to column j of M (that is, to v_j) is 1 if $i = j$ and 0 otherwise. So $\varphi_1 = (1\ \ -1)$, $\varphi_2 = (-1\ \ 2)$. Check: $\varphi_1(v_1) = 2 - 1 = 1$, $\varphi_1(v_2) = 0$, $\varphi_2(v_1) = -2 + 2 = 0$, $\varphi_2(v_2) = 1$.
3. The coordinates of ψ in the dual basis are its values on the basis: $\psi(v_1) = 7$, $\psi(v_2) = 4$, so
$$\psi = 7\varphi_1 + 4\varphi_2.$$
Check: $7(1\ \ -1) + 4(-1\ \ 2) = (3\ \ 1)$.